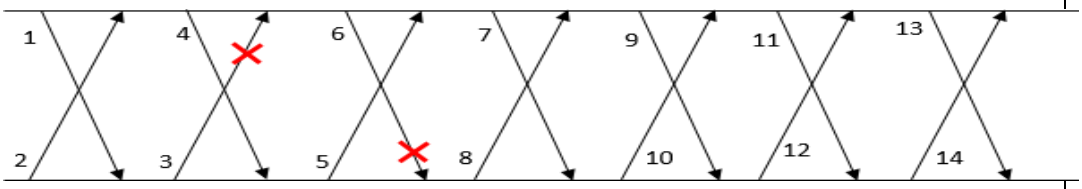
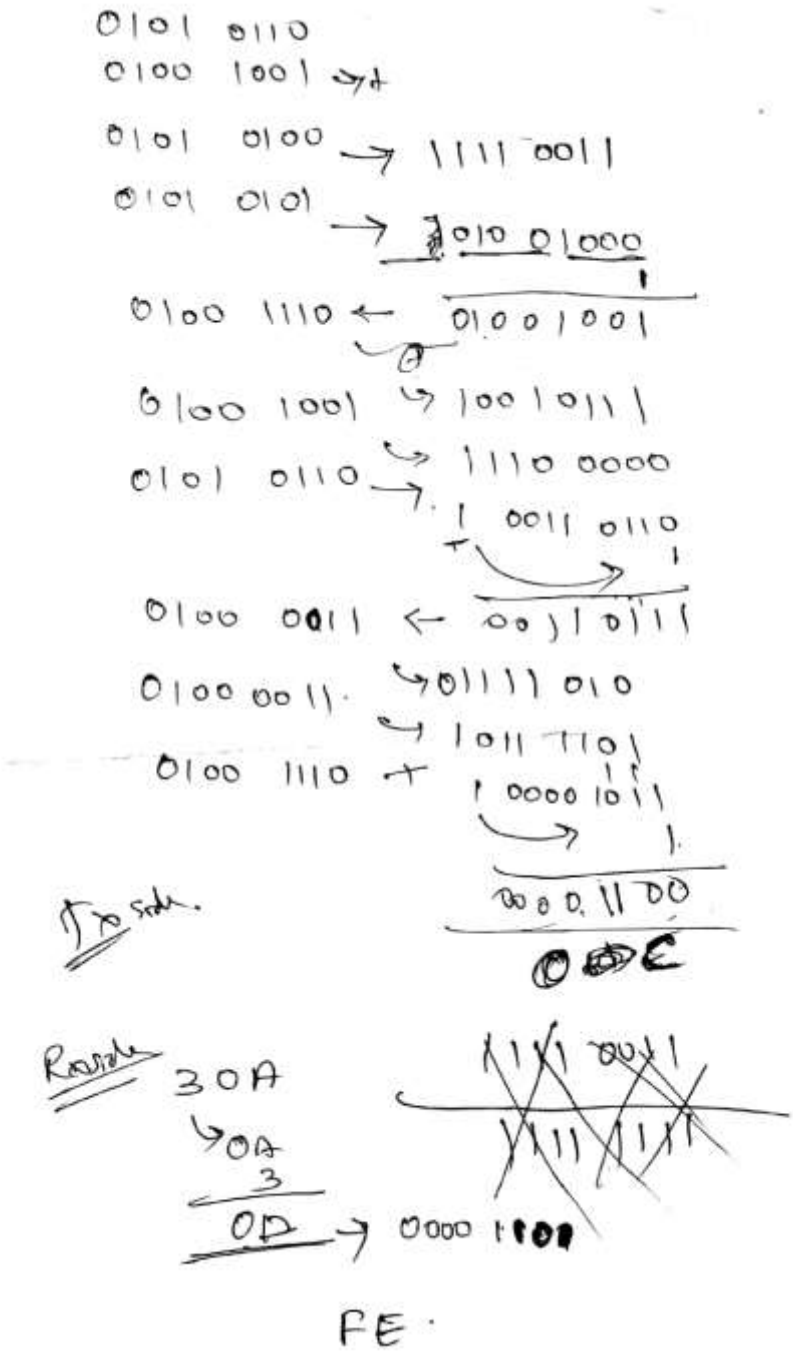
School of Electronics Engineering
Fall Semester 2023-24
Continuous Assessment Test - II

Programme : B.Tech (ECE)
Course Code : BECE401L Course Name : COMPUTER COMMUNICATIONS AND NETWORKS
Faculty Name : P PRAKASAM
Slot : C1+TC1/C2+TC2 Class ID : VL2023240100284/VL2023240100285
Date of Exam : 01.07.2023

Max 50 Marks

Q.No	Question	Max Marks	CO	BL
1	An IT industry has been assigned with IP address 165.15.0.0. It requires two hundred and fifty subnetworks and each with two hundred and fifty hosts. Find the following to configure the subnet. a) Default subnet mask 255.255.0.0 b) Custom Subnet mask 255.255.255.0 c) Number of bits borrowed 8 d) Number of usable host addresses 254 e) Number of usable subnets 254 f) Subnet id for 200th subnet 165.15.200.0 g) First host address for 101th Subnet 165.15.101.1 h) Broadcast address for 100th Subnet 165.15.100.2550 i) Last valid host address for 50th subnet 165.15.50.254 j) Valid address ranges for 49th subnet 165.15.49.1 - 165.15.49.254	10	CO 4	BL5
2	(i). Write the shortest form of the following addresses. a. 2340: lABC:119A:A000:0000:0000:0000:0000 2340:1ABC:119A:A000:: b. 0000:00AA:0000:0000:0000:0000: 119A:A231 0:AA::119A:A231 c. 2340:0000:0000:0000:0000: 119A:A001:0000 2340::119A:A001:0 d. 0000:0000:0000:2340:0000:0000:0000:0000 0:0:0:2340:: e. 00CB:0000:0000:00AA:0000:0000:0000:CCCC CB:0:0:AA::CCCC (ii). What is the type of each of the following addresses? a. 0::0 Loop Back/Embedded IPv4 b. 0: :FFFF:0:0 Loop Back/Embedded IPv4 c. 582F:1234::2222 Reserved for Future d. 4821::14:22 Reserved for Future e. 54EF::A234:2 Reserved for Future	10	CO 4	BL3
3	The following corresponds to an HDLC ABM frame exchange. Consider 2 bits to represent the sequence number. Complete the diagram by completing the labeling of the frame exchanges.  1) BI00 2) AI00 3) AI11 4) BI11 5) AI22 6) BI21 7) AREJ1 8) BREJ2 9) BI21 10) AI12 11) BI32 12) AI23 13) BI03 14) AI30	10	CO 3	BL5

4	i). In the data link layer of a computer network, bit-oriented framing and bit stuffing method is applied. Assuming that 011110 bit string is used as the starting and ending delimiter. a). At the sender, if the bit string delivered by the network layer to the data link layer is given as follows, obtain the bit string at the data field of the sender's data link layer. 1 0 1 1 1 1 1 1 0 0 1 1 1 0 0 0 1 1 1 1 011110 1 0 1 1 1 0 1 1 1 0 0 0 1 1 1 0 0 0 0 1 1 1 0 1 011110 b). At the receiver, if the bit string delivered by layer 1 to layer 2 is as follows, obtain the bit string before bit stuffing. 0 1 1 11 0 0 0 1 0 1 0 0 1 110 0 11 1 0 11 1 0 0 0 1 11 0 0 1 11 0 1 0 0 0 1 1 11 0 ~~0 1 1 11 0~~ 0 0 1 0 1 0 0 1 11~~0~~ 0 11 1 ~~0~~ 11 1 ~~0~~ 0 0 1 11 ~~0~~ 0 1 11 ~~0~~ 1 0 0 ~~0 1 1 11 0~~ 0010100110111100110111 00 ii). A sender needs to send VITUNIVCCN to a receiver. Find the checksum at the sender site and the receiver site for the following case due to channel error: U is detected as V Hint: ASCII Code is represented in hexadecimal for 'V' is 56, 'I' is 49, 'T' is 54, 'U' is 55, 'N' is 4E and 'C' is 43. 	5+5	CO 3	BL3

5	A telephone modem is used to connect a personal computer to a host computer. The speed of the modem is 64 kbps and the reacting time is 60 ms. Assume ACK frame and overhead frames are 10 bytes. Find the efficiency for Stop-and-Wait, Go-Back N and Selective Reject ARQ schemes if the frame size is 1024 bytes for the bit error rate of 0, 10^{-5} and 10^{-3}. Comment on your results.	10	CO 3	BL3

Error Rate	0	10^{-5}	10^{-3}
S&W	66.96%	61.67%	0.0184%
GBN	99.02%	87.35%	0.014%
SR	99.02%	91.1%	2.73%

Selective Repeat outperforms GBN and S&W, especially at high error rates. The efficiency decrease of SR at high error rates is inevitable for any ARQ protocol